

Question 1 : What is Data Loading in Power BI and why is it considered the first step of analysis?

Ans: In Power BI, Data Loading is the process of importing data from different sources (like Excel, databases, or cloud services) into Power BI so that it can be used for analysis and reporting.

What is Data Loading?

- It is the step where raw data is brought into Power BI from external sources
- Data can be:
 - Imported into the model
 - Or connected via Direct Query (live connection)

Why is it the first step of analysis?

1 Without data, analysis cannot begin

- You need data loaded before you can transform, model, or visualize it

2 Foundation for all further steps

- Data cleaning, modeling, and visualization all depend on loaded data

3 Centralized data access

- Combines data from multiple sources into one place

4 Ensures data availability

- Makes data ready for transformation and reporting

Question 2 : Explain the difference between “Load” and “Transform Data” in Power BI?

Ans: In Power BI, the options Load and Transform Data serve different purposes when bringing data into the system:

 Load

- Directly imports data into Power BI without changes
- Data goes straight into the data model
- Quick and simple
- Used when data is already clean

Think of it as: “Bring data as it is”

Transform Data

- Opens Power Query Editor
- Allows you to clean and shape data before loading
- You can:
 - Remove duplicates
 - Change data types
 - Filter rows
 - Split or merge columns

Think of it as: “Clean and prepare data before using it”

Question 3 : What is a Fact Table and a Dimension Table? Give examples from the dataset?

Ans: In **Power BI**, data modeling commonly uses **Fact Tables** and **Dimension Tables** to organize data efficiently.

Fact Table

Definition:

- Stores **quantitative, measurable data** (numbers)
- Represents **transactions or events**

Characteristics:

- Contains numeric values (e.g., sales, price, quantity)
- Includes foreign keys linking to dimension tables

Example (from a real estate dataset):

- Sale Price
- Property ID
- Transaction Date
- Area/Size
- Number of Rooms

This table answers: “**What happened?**”

Dimension Table

Definition:

- Stores **descriptive attributes** (context/details)
- Used to **categorize and filter data**

Characteristics:

- Contains text or descriptive fields
- Helps in grouping and filtering

Example (from a real estate dataset):

- Property Type (Apartment, House, Villa)
- Location (City, Area)
- Customer Name
- Builder/Developer
- Date Details (Year, Month)

This table answers: **“Who, what, where, when?”**

Relationship Example:

- **Fact Table:** Sales data
- **Dimension Tables:** Property, Location, Customer, Date
- Linked using keys like Property ID or Date ID

Simple interview answer:

“A Fact Table stores numerical transaction data like sale price and quantity, while Dimension Tables store descriptive data like property type and location.”

Question 4 : Why is Star Schema preferred over Snowflake Schema in Power BI?

Ans: In Power BI, the Star Schema is generally preferred over the Snowflake Schema for several important reasons:

1 Simpler Structure

- Star Schema has:
 - One central Fact Table
 - Directly connected Dimension Tables
- Easier to understand and manage
Snowflake schema has more complex, normalized tables

2 Better Performance

- Fewer joins between tables
- Faster query execution and report loading
- Snowflake schema requires more joins → slower performance

3 Improved Report Performance

- Faster visuals and dashboards
- Efficient filtering and aggregation

4 Easier to Use

- Simple relationships
- Easier for beginners and business users to work with

5 Optimized for Power BI

- Works well with Power BI's in-memory engine
- Aligns with best practices for modeling

Practical Questions

- **Question 5 : Identify and remove duplicate records based on Date, Country, and State.**
- **Question 6 : Identify and replace null values in.**
- **Question 7 : Create a new column to calculate Recovery Rate.**

100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty
158		41424		123788		10288	No		13600.00%
1977		44002		98239		17709	No		7300.00%
956		53744		27928		17543	Yes		11400.00%
2071		4795		281466		4650	Not Vaccinated		8200.00%
403		10634		451930		4987	Not Vaccinated		4200.00%
4611		4798		351040		15200	Yes		9300.00%
2779		22420		63292		4697	No		35300.00%
1387		51507		30055		16724	Not Vaccinated		9300.00%
3453		55039		306035		3828	No		13700.00%
955		29300		82912		18352	Yes		200.00%
1712		24345		138660		17087	No		5600.00%
4703		10749		353680		12605	Not Vaccinated		11100.00%
1273		38595		269448		5482	Not Vaccinated		7300.00%

PROPERTIES

Name
coronavirus_2020_dataset

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Changed Type with Locale
- Removed Duplicates
- Removed Duplicates1
- Replaced Value
- Added Custom
- Rounded Off
- Changed Type2
- Changed Type1

- Question 8 : Create a summarized table showing total confirmed cases by Country.

Country	Sum of Confirmed_Cases
Brazil	4804531
India	4198680
Italy	3390760
Spain	4354613
UK	3651598
USA	3399318
Total	23799500